
Refinement Buys Intelligibility, Search Buys Identity: Per-Metric Test-Time Scaling for Masked-Diffusion TTS

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Abstract

1 A diffusion language model buys sequential computation twice: depth, paid once
2 in parameters, and denoising steps, rented per utterance at serve time. We ask
3 whether the rented kind helps every capability equally. We train an iso-parameter
4 width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 19–133 M
5 non-embedding parameters) and sweep the refinement budget T at inference only,
6 giving a 75-point (w, d, T) surface per metric from 45 trained models. We fit the
7 pre-registered form $\text{err} = E + Aw^{-\alpha} + Bd^{-\beta} + CT^{-\tau}$ to that surface. Test-time
8 scaling is metric-selective: $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$, 95% run-level bootstrap
9 CI [0.092, 0.131], excluding zero. The selectivity is a difference in *magnitude*, not
10 in convergence *rate*. Over $T=1 \rightarrow 16$ refinement moves 0.952 absolute WER
11 (a $6.1\times$ reduction) but only 0.161 absolute speaker-similarity error ($1.25\times$), a
12 $5.9\times$ asymmetry; on each metric’s own normalised curve, $T=8$ realises 94.6%
13 of the total WER gain and a marginally smaller 93.7% of the total identity gain.
14 The selectivity persists across a 276 M four-budget replication, $\Delta\tau = +0.1095$,
15 CI [0.0951, 0.1241]. The limit is on what *refinement* buys rather than on what
16 inference buys: at matched NFE, best-of- K search moves speaker similarity by
17 $+0.0444$ (CI [0.0416, 0.0470]) over refinement at $T=64$, and by $+0.0365$ over the
18 deployable $T=16$ default, where refinement moves it least. Depth and steps are *not*
19 interchangeable: the substitution form $B(dT^\kappa)^{-\beta}$ fits worse than the separable
20 form by $\Delta\text{AICc} = +69.3$.

21 1 Introduction

22 A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer
23 blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps:
24 T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An
25 autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the
26 moment training ends. If rented serial compute substitutes for owned serial compute, a deployment
27 can trade parameters for steps after the fact. That trade is available only for the capabilities that steps
28 improve.

29 Zero-shot text-to-speech is a clean probe: intelligibility is compositional (the right phonetic content in
30 the right order, read out by word error rate) while speaker identity is representational (timbre carried
31 from a short prompt, read out by SIM-o). The two need not respond to the same knob.

32 We train one iso-parameter width–depth grid of masked-diffusion codec language models and then
33 sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional
34 training cost. Four hypotheses were pre-registered before the first run. Two are primary: refinement
35 steps improve intelligibility more than identity, so that test-time scaling is metric-selective (H-D2);

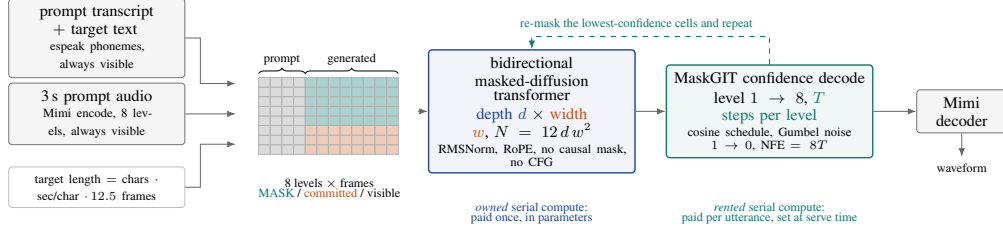


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes and a 3-second prompt are always visible; the generated region of the $8 \times \text{frames}$ token grid starts fully **masked**. The backbone spends **depth** — *owned* serial compute, fixed at training time — and the sampler spends **T refinement steps per codebook level** — *rented* serial compute, chosen at serve time. One training grid over (w, d) therefore yields a whole (w, d, T) surface at inference. This paper asks whether the rented kind helps every capability equally.

and, for intelligibility, steps and depth exchange at a measurable rate κ , the number of step-doublings that buys one depth-doubling (H-D3). Two are secondary: at fixed T , iso-parameter shape matters beyond parameter count and the two capabilities differ in their width–depth anisotropy (H-D1); and fits on the two smaller parameter budgets predict the largest budget (H-D4). Eq. (1) states the two fitted forms: one separable in width, depth and steps, one in which steps substitute for depth through the product dT^κ .

$$\begin{aligned} \text{separable: } \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \\ \text{substitution: } \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m (dT^\kappa)^{-\beta_m} \end{aligned} \quad (1)$$

Steps are metric-selective: $\Delta\tau = 0.110$, run-level bootstrap CI [0.092, 0.131], excluding zero. Depth and steps do not exchange. The substitution form fits decisively *worse* than the separable one ($\Delta\text{AICc} = +69.3$ for WER, $+71.3$ for SIM-o), and κ runs to its bound, so no finite depth–step exchange rate describes this regime. Extra inference compute is a targeted instrument, not a substitute for architecture.

Contributions. ❶ The first per-metric (w, d, T) scaling surface for speech generation: 75 points per metric from 45 trained models, released with the training and analysis code. ❷ Refinement steps are metric-selective, $\Delta\tau = 0.110$ with CI [0.092, 0.131]. The result holds across four parameter budgets, $3\times$ training compute, a second ASR and an unbounded reparameterisation. Depth–step substitution is refuted on AICc, and κ is unidentified. ❸ A per-axis account of what moves speaker identity, measured against a codec ceiling: training compute and test-time selection move it where refinement does not, while longer prompt context and rate matching buy nothing.

2 Related work

Masked diffusion language models are trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin et al., 2021], but their scaling is reported against a single likelihood-shaped objective, which cannot expose a per-capability difference. The masked-generative recipes that made codec-token TTS practical [Borsos et al., 2023, Chang et al., 2022, Wang et al., 2024] fix a step budget by engineering judgement and report one operating point; none maps error against steps as a law. Autoregressive codec TTS [Wang et al., 2023] has been scaled, but in total parameters only [Ye et al., 2025], which separates neither width from depth nor either from steps. Classical scaling work aggregates the same way [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2024], where the payoff is a single correctness metric. Whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech.

3 Method

Backbone and objective. The backbone is a bidirectional (non-causal) pre-LN transformer over Mimi [Défossez et al., 2024] codec tokens at 12.5 Hz, using the first 8 residual codebooks (2048 entries each, plus MASK and PAD). It is parameterised by **width** w and **depth** d , with head dimension

64, GELU MLP ratio 4, RMSNorm, RoPE and no weight tying, so the non-embedding parameter count is $N = 12 d w^2$. One embedding table per codebook is summed at the input and one linear head per codebook reads out. Training follows SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example we draw a level $l \sim \mathcal{U}\{1..8\}$ and a ratio $t \sim \mathcal{U}(0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , and levels above l are entirely MASK. The loss is cross-entropy on the masked cells of level l only. A phoneme prefix and a 3-second prompt of codec tokens are always visible. The model has no speaker encoder, so the backbone is the only identity pathway, and it is trained without classifier-free guidance, which would otherwise confound the step axis. The target length is fixed before generation from the text length and a corpus character rate (0.0602 s/char).

Frozen sampler. Inference is MaskGIT-style confidence decoding [Chang et al., 2022], run level by level with exactly T refinement steps per level, so NFE = $8T$. A cosine schedule sets how many cells remain masked after each step, tokens are drawn at temperature 1.0, and confidences are log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$; committed cells stay committed. The sampler is identical for every config, seed and T , and generation noise is keyed so that T is the only difference between surface points of one run.

Design and statistics. The grid holds 15 configs: three iso- N budgets $\times$ five aspect ratios, each trained at 3 seeds for 30,000 steps at effective batch 256 with μP [Yang et al., 2021] (base width 256, base LR 0.004, transfer checked by five-point LR sweeps). We sweep $T \in \{1, 2, 4, 8, 16\}$ at evaluation only, which gives 75 surface points per metric from 45 trained models. The metric stack is WER (Whisper-large-v3 [Radford et al., 2023], greedy), SIM-o (WavLM-large [Chen et al., 2022] against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-output rule, and UTMOS [Saeki et al., 2022]. The error transforms are $\text{err}_{\text{WER}} = \text{WER}$ and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. We fit both forms of Eq. (1) by weighted non-linear least squares with 32 multi-starts. The intervals on $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ and $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ are percentile intervals from a bootstrap of 2000 replicates that resamples at the *run* level and recomputes every T point. The run level is the right unit because the five T -points of one run share a trained model, so resampling surface points would understate their correlation.

98 4 Results

Refinement steps are metric-selective. Fitted on all 75 surface points, the step exponent for intelligibility exceeds the one for identity: $\tau_{\text{WER}} = 0.836$ (CI [0.825, 0.851]) against $\tau_{\text{SIM}} = 0.726$ (CI [0.709, 0.744]), so $\Delta\tau = 0.110$ with CI [0.092, 0.131], excluding zero. This is the pre-registered primary result (H-D2). The grid resolves shape differences well above seed noise: within each budget the spread of config means at $T=16$ is at least twice the pooled seed standard deviation, so the pre-registered power criterion passed in all three budgets.

The raw surface shows the same asymmetry without any model. Over the grid, $T=1 \rightarrow T=16$ takes WER from 113.8% to 18.7% (a factor of $6.1\times$ on the error scale) while SIM-o moves only $0.209 \rightarrow 0.370$ ($1.25\times$ on $1 - \text{SIM-o}$), an absolute-error asymmetry of $5.9\times$.

The selectivity is not an artefact of scale, training compute or the metric stack (Table 1). A fourth parameter budget (276M, ten further runs) gives $\Delta\tau = +0.1095$, CI [0.0951, 0.1241]; retraining three configurations $3\times$ longer gives $+0.1767$, CI [0.0789, 0.2854]; a different ASR gives $+0.1283$ and unbounded log-amplitudes $+0.1260$. Extending T to 64 leaves WER still falling past the grid edge ($0.1516 \rightarrow 0.1294$), so the gain measured at $T=16$ is a lower bound.

Depth and steps do not substitute. The substitution form of Eq. (1) fits the surface decisively worse than the separable form: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ for WER and $+71.3$ for SIM-o, where the pre-registered claim (H-D3) required $\leq +4$. The exchange rate itself is unidentified: κ sits at its bound for both metrics. The fits show why. T enters as an additive term that decays to a floor, whereas depth moves the floor itself, and a single factor $d T^\kappa$ cannot reproduce both behaviours.

Shape matters, but the anisotropy ratio is not identified. At $T=16$ the two-axis model beats the N -only model for both metrics ($\Delta\text{AICc} = -15.5 / -5.5$ against a -4 threshold), so error at fixed

Table 1: $\Delta\tau$ survives variation in scale, training compute and the measurement stack. Each row refits the surface with one factor changed: a fourth parameter budget, $3\times$ training compute, a different ASR, and an unbounded reparameterisation of the fitted amplitudes. Every interval excludes zero.

variation	what changes	$\Delta\tau$	95% CI
none (75 points, 45 runs)	—	0.110	[0.092, 0.131]
fourth budget, 276 M	10 further runs	+0.1095	[0.0951, 0.1241]
$3\times$ training compute	3 configs at 90k steps	+0.1767	[0.0789, 0.2854]
ASR swapped	Whisper-medium.en	+0.1283	[0.1064, 0.1492]
unbounded amplitudes	$\log A, \log B, \log C$ free	+0.1260	[0.1043, 0.1474]

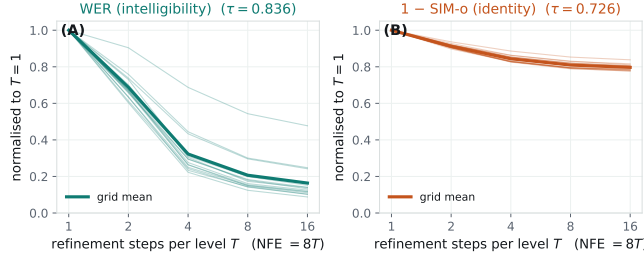


Figure 2: Test-time scaling is metric-selective. Error versus refinement steps for all 15 configs, each normalised to its own $T=1$ value; thick lines are grid means. (A) WER error falls 0.952 in absolute terms over $T=1 \rightarrow 16$ and is still falling at $T=16$. (B) SIM-o error falls 0.161 over the same range, $5.9\times$ less. Same models, same frozen sampler, same items — only T differs.

parameter count depends on how those parameters are shaped. The direction of that dependence does not resolve. The width amplitude saturates its pre-registered upper bound in every fit, so only the product $A w^{-\alpha}$ is identified and α is not. The ratio $\rho = \alpha/\beta$ inherits the problem: $\Delta\rho = 0.174$ with a CI $[-0.711, 0.388]$ that spans zero, so the secondary pre-registered claim about the direction (H-D1) is not supported.

Depth is identified, and it separates the capabilities sharply ($\beta_{\text{WER}} = 2.040$ versus $\beta_{\text{SIM}} = 0.553$): moving from the shallowest shape to the best depth is worth 40.2, 16.3 and 7.3 WER points across the three budgets, while SIM-o varies by under 0.065. The depth optimum is interior only in the smallest budget; in the two larger ones the argmin sits at the deepest shape tested, so those are lower bounds, not interior optima.

Extrapolation and degenerate outputs. Fitting the two smaller budgets and predicting the largest gives MAPE 2.4% for SIM-o against 2.2% for the N -only model: identity extrapolates across scale, within the 15% we pre-registered, but no more accurately than parameter count alone. WER does not extrapolate (20.2% versus 20.4%). Degeneracy falls monotonically with T , from 40.4% at $T=1$ to 0.1% at $T=16$: most of what a one-step budget costs is outright failure, not degraded speech.

Serving corollary. Per-layer latency is flat in width: measured at batch 1, c_{layer} is 0.493–0.514 ms for $w \in [256, 1152]$, because the device is launch-bound at this scale. Serial latency is therefore $8 T d c_{\text{layer}}$, so at fixed parameter count width is free and a latency budget constrains only the product $T d$. How to divide that product between depth and steps is not determined here. The pre-registered rule, buy depth before steps, is refuted: it holds at 5.9% of the 40 latency budgets we test, against a required 80%. The reverse ordering is a boundary effect of a grid censored at $T=16$, not a finding.

What buys identity, measured against a ceiling. If refinement is an intelligibility knob, what moves identity? We put every candidate lever on one scale and read it against a fixed bound. A codec round-trip of ground-truth audio scores 0.555, which leaves 0.0747 of headroom above our best system.

Training compute is the largest lever: 30k $\rightarrow$ 90k steps gains +0.0827 (CI [0.0733, 0.0908] over 9 runs), and a further doubling adds +0.0155. Test-time search is the second lever: best-of- K sampling at matched NFE ($8T$ per candidate), selecting with one speaker encoder and scoring with a second. It gains +0.0444 over refinement at $T=64$ (CI [0.0416, 0.0470], on 72 % of items) and +0.0365 over

Table 2: *Fitted exponents* on the 75-point surface (M_{sep} for α, β, τ ; M_{sub} for κ), with 95% run-level bootstrap intervals on the step exponents. The primary result is $\Delta\tau = 0.110$, CI [0.092, 0.131], which excludes zero. κ sits on its bound in both rows and the substitution form is worse than the separable one by $\Delta\text{AIC}_{\text{sub-sep}} \gg +4$, so the depth-step exchange rate has no finite estimate. The negative $\Delta\text{AIC}_{\text{full-N}}$ says that shape matters at fixed N .

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AIC}_{\text{sub-sep}}$	$\Delta\text{AIC}_{\text{full-N}}$
WER	1.574	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†]at the fitted bound, i.e. not identified; $\Delta\tau = 0.110$, CI [0.092, 0.131].

Table 3: *What buys speaker identity.* Absolute SIM-o gain per axis, set against the 0.0747 of headroom between our best system and a codec round-trip of real audio. Training compute and test-time selection buy the most. Guidance buys a little and charges for it in WER; rate matching and longer prompts buy nothing.

axis	change	Δ SIM-o	cost
training compute	30k $\rightarrow$ 90k steps	+0.0827	3 $\times$ training
(further)	90k $\rightarrow$ 180k	+0.0155	2 $\times$ again
test-time search	best-of-8 vs $T=16$	+0.0365	+0.0425 WER, 8 $\times$ NFE
parameters	20 M $\rightarrow$ 276 M	+0.0531	14 $\times$ parameters
guidance	$\gamma = 0.5$	+0.0121	+2.73 WER points
rate matching	per-item duration	+0.0044	free
prompt context	3 s $\rightarrow$ 9 s	none	+71.7 WER points

the deployable $T=16$ default, at a cost of +0.0425 WER. Refinement remains the better spend for intelligibility: WER is 0.1316 under refinement against 0.1955 under search. The gain comes from selection, not from sampling: an arbitrary candidate scores 0.4259 regardless of K . It persists at 276 M non-embedding parameters (+0.0422) and, on three runs, at 3 $\times$ training compute (+0.0368).

The remaining levers are small or empty. A 14 $\times$ parameter increase gains +0.0531. Training-free speaker-contrastive guidance gains +0.0121 and costs +2.73 WER points, so it trades one capability for the other. Rate-matched length conditioning gains +0.0044, and longer prompts gain nothing: the best prompt is the 3 s default seen in training, and beyond it the output stays fluent and speaker-consistent while decoupling from the text. Identity is bought with training and with selection, not with more refinement.

Where the passes are spent matters as much as how many: at matched NFE = 32, moving them to the coarsest codebook level changes WER from 1.1480 to 0.2191, at a cost in identity and naturalness (Appendix D).

5 Limitations and outlook

Every model in the grid is under-trained. We cap non-embedding parameters at 133 M and training at 30,000 steps, so T_{WER}^* is a lower bound. Part of the identity gap is also not ours to close: a codec round-trip of the target scores 0.555 against 0.678 for the original waveform, so 62% of the distance to real speech is lost in the codec before the model generates anything.

The search result rests on two speaker encoders of shared lineage. Selector and scorer are both ECAPA-TDNN-family models, and their per-item scores correlate at r in the range 0.71–0.73. Two facts bound how much of the measured gain that overlap could account for. Each encoder separates ground-truth speaker pairs on its own, and the selector recovers only 46 % of an oracle selector’s gap, where a shared bias would predict nearly all of it. Neither fact excludes representational overlap; excluding it needs a scorer from a different encoder family. Compute matching covers the diffusion transformer only, so the selector and scorer forward passes are not charged to the search arm.

The study is one language, one codec and one corpus, with no classifier-free guidance. The $\Delta\tau$ interval reflects seed variation only. Target length comes from a corpus median character rate rather than a per-item prediction. Two fitted quantities are not identified: κ sits on its bound in every fit, and the width amplitude saturates its bound, which leaves $\Delta\rho$ undetermined.

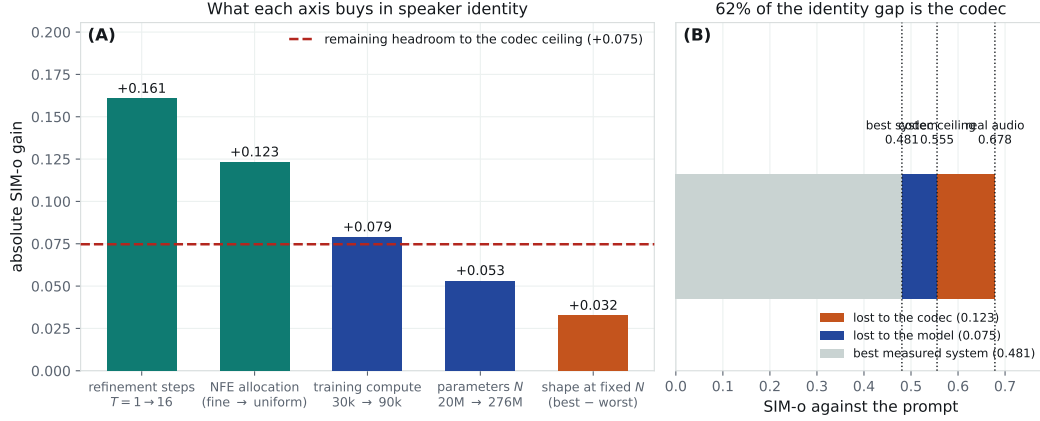


Figure 3: *Identity is mostly not the model's to win.* (A) what each axis buys in speaker similarity, against the headroom that remains to the codec ceiling. (B) of the distance between real audio and our best system, 62% is lost in the codec itself.

Table 4: *Refinement and search buy different capabilities at the same cost.* Matched NFE (8T per candidate); best-of- K candidates are selected by WavLM-SV and scored by a separate ECAPA encoder. Refinement wins intelligibility at every tier, and search wins identity at every tier.

NFE	refinement	search	Δ ECAPA	95% CI	WER (refine / search)
128	$T=16$	$K=2$	+0.0153	[0.0127, 0.0179]	0.1530 / 0.1938
256	$T=32$	$K=4$	+0.0322	[0.0307, 0.0338]	0.1335 / 0.1947
512	$T=64$	$K=8$	+0.0444	[0.0416, 0.0470]	0.1316 / 0.1955
replicated at NFE 512 outside the grid: 276 M budget +0.0422 [0.0393, 0.0448]; $3 \times$ training +0.0368 [0.0321, 0.0428]					

The analysis ran as pre-registered, with none of the declared analyses dropped, and the result falls in outcome class S1, which the protocol reserved for a positive $\Delta\tau$ whose interval excludes zero. Two censorings remain, and both are removable by a larger grid: the depth optimum sits at the deepest shape tested in the two larger budgets, and WER is still falling at the top of the fitted T range. Running deeper and further in T would locate both optima and let the owned and rented levers be compared where each has converged.

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Table 5: *Where the refinement passes are spent matters as much as how many.* Three frozen per-level schedules at matched total NFE = 32 on the 125 M budget, 15 runs, 400 items. Spending every pass on the coarsest codebook level rather than the finest cuts WER by $5.2\times$, and costs identity and naturalness against a uniform split.

schedule	WER	SIM-o	UTMOS	degenerate
coarse [25, 1, . . . , 1]	0.2191	0.3057	2.17	0.1%
uniform [4, . . . , 4]	0.3144	0.3477	2.51	0.5%
fine [1, . . . , 1, 25]	1.1480	0.2246	1.56	39.1%

Table 6: *Longer prompts do not buy identity, and destroy intelligibility.* Prompt length swept at inference on 15 125 M-budget runs, 397 items, every arm scored against the same fixed 3 s reference. Training used 3 s prompts only, so the 6 and 9 s arms test extrapolation. There the output stays fluent and speaker-consistent, but it decouples from the conditioning text.

prompt	SIM-o	WER
1.5 s	0.3213	0.1578
3 s (<i>training length</i>)	0.3839	0.1474
6 s	0.3724	0.7692
9 s	0.3759	0.8642

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238 A Reproducibility

239 Protocol, hypotheses, thresholds, the frozen sampler and the analysis plan were registered before the
 240 first run, and every gate is logged with its measured values. The released artefact holds the harness,
 241 all 45 run records, the 75-point surface, and the fitting and bootstrap code. Total cost 143 GPU-hours.

242 B The shape grid, as run

243 Every config in the active grid with its measured $T=16$ means (seeds 0, 1, 2, 400 eval_zs items each).
 244 Iso- N within a budget: the three budgets sit at 20, 50 and 125 M non-embedding parameters and the
 245 five shapes per budget span the aspect-ratio range at fixed $N = 12dw^2$.

	config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	Degen (%)
	A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
	A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
	A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
	A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
	A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
	B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
246	B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
	B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
	B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
	B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
	C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
	C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
	C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
	C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
	C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

247 C Additional figures

248 D Refinement-budget allocation across codebook levels

249 Three frozen per-level step schedules at matched total NFE = 32 on the 125 M budget (15 runs, 400
 250 items). Paired item-level bootstrap, 2000 replicates, pairing on the eval item so item difficulty cancels.
 251 Coarse spends all passes on codebook level 0, fine on level 7, uniform splits evenly.

	schedule	WER	SIM-o	UTMOS	degenerate
252	coarse [25, 1, ..., 1]	0.2191	0.3057	2.17	0.1%
	uniform [4, ..., 4]	0.3144	0.3477	2.51	0.5%
	fine [1, ..., 1, 25]	1.1480	0.2246	1.56	39.1%

253 Allocation is metric-selective in the same way the step *count* is: coarse wins WER but loses identity
 254 and naturalness against uniform, both with intervals excluding zero. It is a trade, not a free win.

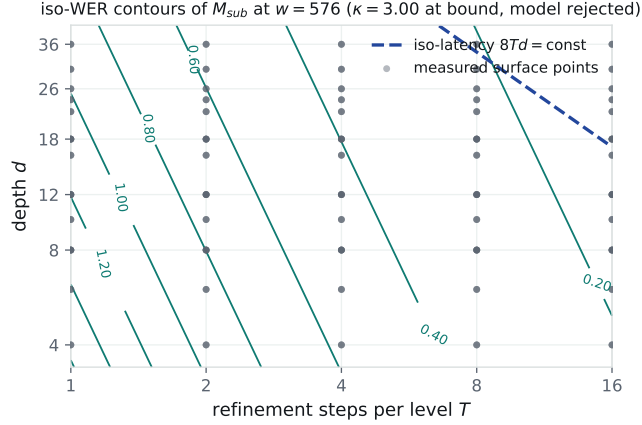


Figure 4: *Why the substitution form is rejected.* Iso-WER contours of M_{sub} in $(\log T, \log d)$ at the median width, with the iso-latency line $8Td = \text{const}$. The fitted κ sits on its bound and $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$, so these contours are shown as a rejected model, not as a design rule.

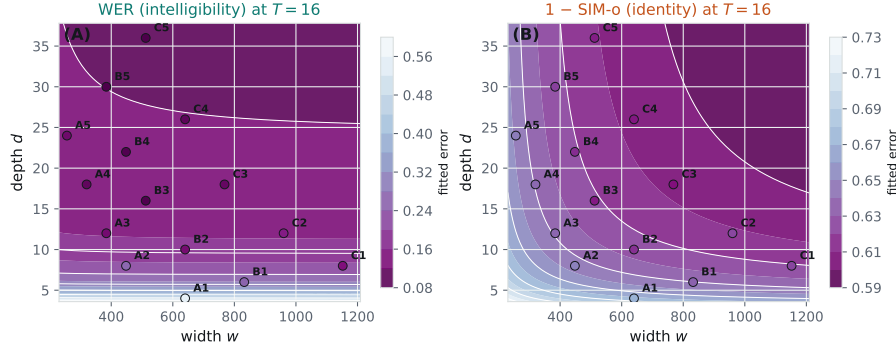


Figure 5: *Fitted anisotropy surface at $T=16$ for both metrics, with the measured config means overlaid.* The width amplitude A saturates its pre-registered upper bound for both metrics, which is why $\rho = \alpha/\beta$ is reported as unidentified in §4 even though the two-axis model beats the N -only model ($\Delta\text{AICc} = -15.5$ and -5.5).

255 **A significance-based saturation contrast.** Defining saturation as the smallest T whose paired-
 256 bootstrap interval for $\text{err}(T) - \text{err}(64)$ covers zero gives $T_{\text{WER}}^* = 32$ against $T_{\text{SIM}}^* = 16$ on the
 257 extended grid. We report this only here and do not build on it: roughly half the contrast is a precision
 258 artefact, since identity’s detection floor is $\approx 1.8\times$ looser relative to its own dynamic range than
 259 intelligibility’s. The raw-scale T^* — 95% of the $T=16$ value — is not affine-invariant and must
 260 never be compared across metrics: the same 5% band is 0.67% of WER’s range but 11.96% of
 261 SIM-o’s. On the invariant fraction-of-total-gain definition, $T^* = 16$ for both metrics.

262 E Fit diagnostics and exploratory analyses

263 **Parameters at bounds.** The width amplitude A saturates the pre-registered *upper* bound $A \leq 10$ in
 264 every fit (Part A: 10.000 for both metrics; Part B: 9.999 for WER, 10.000 for SIM-o), so only the
 265 product $A w^{-\alpha}$ is identified and α is not; the bound, not the data, is binding, which is a limitation
 266 of the pre-registered parameterisation worth fixing in any follow-up. κ reaches its upper bound in
 267 every one of the 2000 bootstrap replicates of M_{sub} . The irreducible term E reaches 0 for WER under
 268 M_{sep} . The depth amplitude B is interior throughout, so β is identified.

269 **Exploratory: the depth exponent separates the capabilities.** $\beta_{\text{WER}} = 2.040$ against $\beta_{\text{SIM}} =$
 270 0.553 , a factor of 3.7 — the asymmetry H-D1 sought, in the parameter the data pin down. Part-B ρ

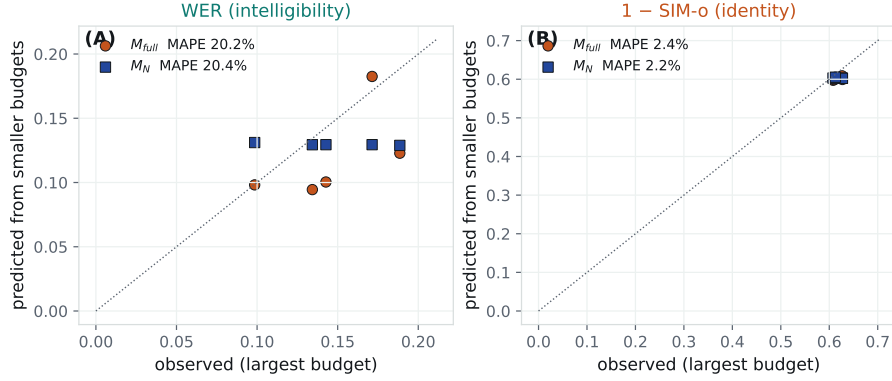


Figure 6: *H-D4 extrapolation*. The two smaller budgets are fitted and the largest budget predicted, under M_{full} and the N -only model. SIM-o extrapolates (MAPE 2.4 % vs 2.2 %); WER does not (20.2 % vs 20.4 %).

values point the pre-registered way ($\Delta\rho = +0.890$) but inherit the A -cap problem; the pre-registered test is Part A, so this is exploratory only.

Weighting. Points are weighted $1/SE^2$ with SE taken over seeds, floored at a quarter of the pooled seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored points is recorded in the released `fits.json`.

Exploratory (not part of any declared claim). An unweighted refit of both parts is released alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed. Both reproduce the sign and significance of $\Delta\tau$.

Use of Generative AI Disclosure

Generative AI was used as a coding assistant for the experimental harness and for language refinement of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and take full responsibility for this paper.